K-EL/01 Simulation & Verification Plan

Proposed validation protocol — no analyses or physical tests are represented as completed

1. How to use this plan

The original challenge explicitly does not require detailed FEA, manufacturing drawings, component purchases or a physical prototype. The plan below is a staged future verification route. Execute only after required mission inputs and supplier configuration are available; retain raw evidence and record failures, not just successful plots.

ID	Verification	Scope	Owner	Evidence	Acceptance logic
V-01	Requirements review	Duty profile, speed range, accelerations, accuracy/backlash, impacts, temperatures, life, power-loss hold.	Project lead and engineering	Signed requirement/assumption register	All key unknowns answered or formally bounded before design freeze.
V-02	Static torque audit	Horizontal-arm torque and all load directions/ranges; confirm mass/COM includes links/attachments.	Mechanical engineer	Calculation workbook and independent check	Numerical results reconcile; input sources and SI units explicit.
V-03	Drive operating point	Motor torque-speed curve at actual voltage; gear ratio/input speed; loaded efficiency; continuous/intermittent output torque.	Supplier + controls engineer	Configured part-number sheet and curves	Output 45 RPM and target torque achieved simultaneously at defined duty.
V-04	Duty/thermal model	Trajectory and reflected inertia; peak/RMS torque; winding and gearhead heating; temperature rise and derating.	Controls/mechanical engineer	Torque profile, loss model, thermal assumptions	Predicted temperatures below manufacturer limits with defined margin.
V-05	Coupling interface	Hub bore/keyway, shaft diameters, torque reversal, speed, offset/angular misalignment, torsional stiffness.	Mechanical engineer + KTR	Selected hub/spider codes and interface drawing	All torque, speed and misalignment limits satisfied including service factor.
V-06	Spindle strength	Bending+torsion, keyway notch, fatigue, shoulders, retaining grooves, deflection, stress concentrations.	Mechanical engineer	Detailed shaft CAD/drawing and checked report	Margin per agreed design standard and peak/duty load cases; no reliance on smooth-shaft screen alone.
V-07	Bearing and carrier	Radial/axial reactions, L10 and oscillation factors, static safety, fits, lubrication, housing stiffness, fasteners.	Mechanical engineer	Bearing tool output, fit table, structural calculations	Life/stiffness/fit requirements met; vendor constraints observed.
V-08	CAD interference & motion	Full assembly clearance, coupling envelope, fastener access, cable bend, arm sweep, stops, collision.	CAD designer	Native assembly, screenshots, interference report	No unplanned interference over specified swept envelope; maintainability demonstrated.
V-09	Finite-element analysis (optional)	Yoke, flange/root plate and spindle only where hand calculations indicate value. Include realistic contacts, preload, load paths and mesh sensitivity.	FEA-capable engineer	Solver file, mesh convergence, BC/load rationale, stress/displacement plots	Convergence and sensitivity accepted; FEA does not replace bearing, fatigue, or supplier checks.
V-10	Control/safety review	Encoder resolution/placement, polarity, controller limits, emergency stop, regen, brake and fault response.	Controls/safety engineer	Block diagram, configuration, hazard review	Safe state defined; power-loss holding behavior is explicitly met or operation externally supported.
V-11	Bench torque/speed test	Guarded restrained fixture; measure torque and RPM at several duty points and current.	Test engineer	Calibration records, raw CSV, protocol	Performance reaches requirements at actual configured operating point.
V-12	Thermal/duty test	Run representative movement cycle to steady state; monitor winding proxy, casing, gearhead, controller, current and speed.	Test engineer	Raw time series and uncertainty statement	Temperatures stabilize within ratings and current limits for stated duty.
V-13	Backlash/deflection	Apply bidirectional load	Test engineer	Indicator/encoder logs and	Measured reversal error

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		steps; measure output angle/hub displacement and elastic deflection.		test report	and stiffness meet the explicit position-error budget.
V-14	Final review / release	Reconcile deviations, actions, BOM/order codes, drawings, software config and test results.	Design owner + reviewer	Signed design review record	All requirements passed/waived with accountable owner; no open critical hazard.

2. Minimal later bench setup

- Rigid guarded base and restrained arm/fixture; no person under or in plane of suspended arm/payload.
- Calibrated torque transducer or lever/force arrangement with known uncertainty, RPM measurement, current/voltage logging and temperature sensors.
- E-stop, current limits, mechanical travel stops, cable restraint and safe power-disconnect/regeneration management.
- Defined test matrix: unloaded run, low-load direction checks, several static torque points, 45-RPM loaded point, representative trajectory and reversal/backlash measurement.
- Predefine abort limits for current, vibration, abnormal noise, temperature, unexpected movement and fixture displacement.

3. Current evidence versus planned evidence

Available now	Not yet available / must not be claimed
Static calculation from brief inputs; explicit sensitivity analysis; component reference list; valid STEP geometry check; conceptual load/reaction/stress arithmetic.	Supplier-confirmed loaded point, physical torque/speed measurements, thermal steady-state data, measured backlash, detailed FEA, bearing life validation, shaft fatigue assessment, safety certification.
Concept architecture and artwork; BOM and registers identify TBDs.	Manufacturing tolerance stack, full fastener/interface calculations, verified power-loss hold solution, test data from actual components.